D13 The GLayout Visionaries

IEEE Solid-State Circuits Society
Technical Committee on the Open Source Ecosystem (TC-OSE)

Team D13

Analog Comparator for Ambient Light Detection in Wearable Assistive Devices for Visually Impaired Patients

Discord name	Affiliation	Experience	Role
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The Problem — 285 Million People Need Our Solution

Visually impaired individuals face daily hazards from sudden light transitions

285 Million

People visually impaired worldwide

39 million fully blind — WHO World Report on Vision, 2019

60–80%

Of accidents occur during sudden dark-to-light transitions

Corridors, tunnels, outdoor entry — no warning, no time to react

Current Limitations

- Existing aids require manual activation — no real-time sensing
- Commercial sensors are bulky, high power (>1 mW), not wearable
- No always-on, coin-battery powered, CMOS-integrated solution exists

Our Solution

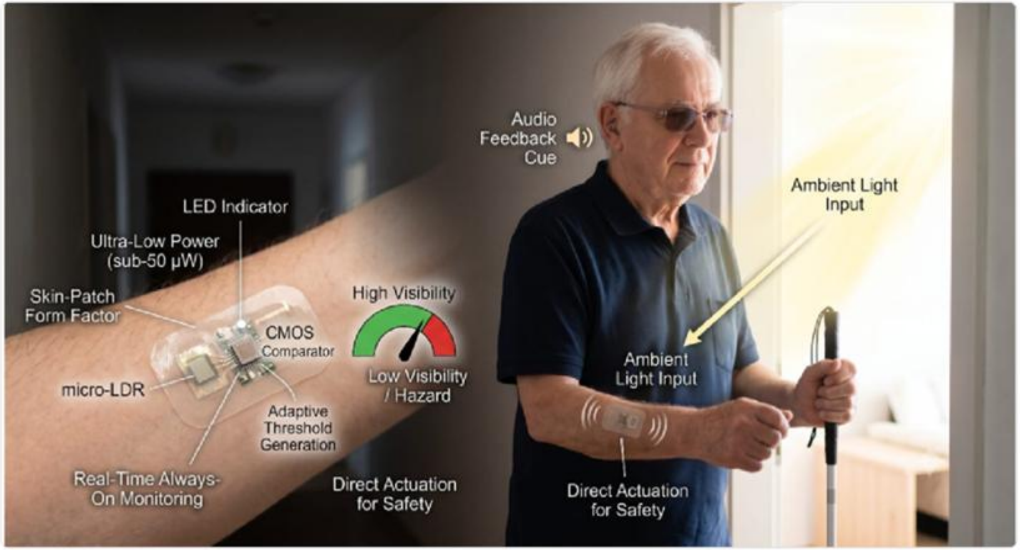
A fully integrated 9-transistor CMOS analog comparator in GF180MCU 180nm PDK — worn as a skin patch — that automatically detects ambient light transitions and triggers haptic or audio alerts in real time.

Why a Comparator?

A comparator converts the analog photocurrent from the photodiode into a clean digital HIGH/LOW decision signal. It is the simplest, lowest-power circuit that can reliably detect light vs. dark transitions without any microprocessor or software.

Application — Real Life Impact for Visually Impaired People

Wearable skin-patch device providing always-on ambient light detection and instant safety alerts



* **Skin-Patch Form Factor**
  Worn on the wrist or forearm — always on, completely passive during normal operation

* **Real-Time Light Detection**
  micro-LDR photodiode captures ambient light and generates a proportional voltage (0–1.8V)

* **Instant Safety Alert**
  Audio beep or haptic vibration triggers within 1 us when light drops below safe threshold

* **Ultra-Low Power**
  Sub-50 uW total power — CR2032 coin battery lasts 1 year+ without recharging

Target scenario: Detecting dark corridors, tunnel entries, sudden nightfall — giving 1us early warning for safety

System Architecture — Signal Processing Chain

From ambient light detection to digital safety alert — five functional stages

1 Light Sensing — Photodiode

External LDR/photodiode converts ambient light to resistance; skin-patch captures Vpatch (0 to 2.8V)

2 Reference Generation — M6, M7

Resistive voltage divider (M6 PMOS + M7 NMOS) sets Vth = 3.06V as the detection threshold

3 Differential Comparison — M1 to M5

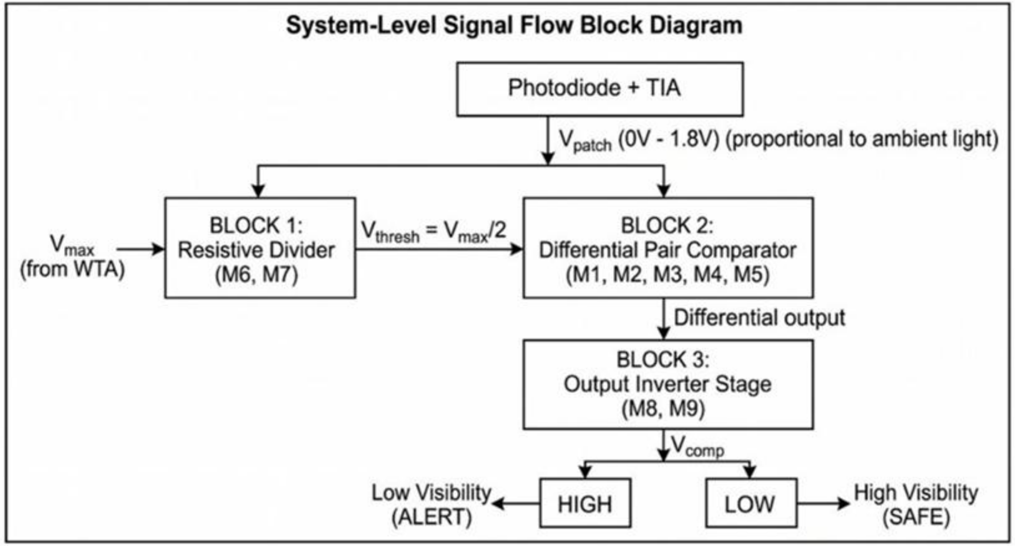
NMOS differential pair (M1, M2) with PMOS active load (M3, M4) and tail current source M5

4 Output Buffering — M8, M9

CMOS inverter stage amplifies differential output to rail-to-rail digital signal

5 Alert Output

Vout HIGH drives haptic/audio alert — response time under 1 us from threshold crossing



Complete Signal Path — Light to Safety Alert

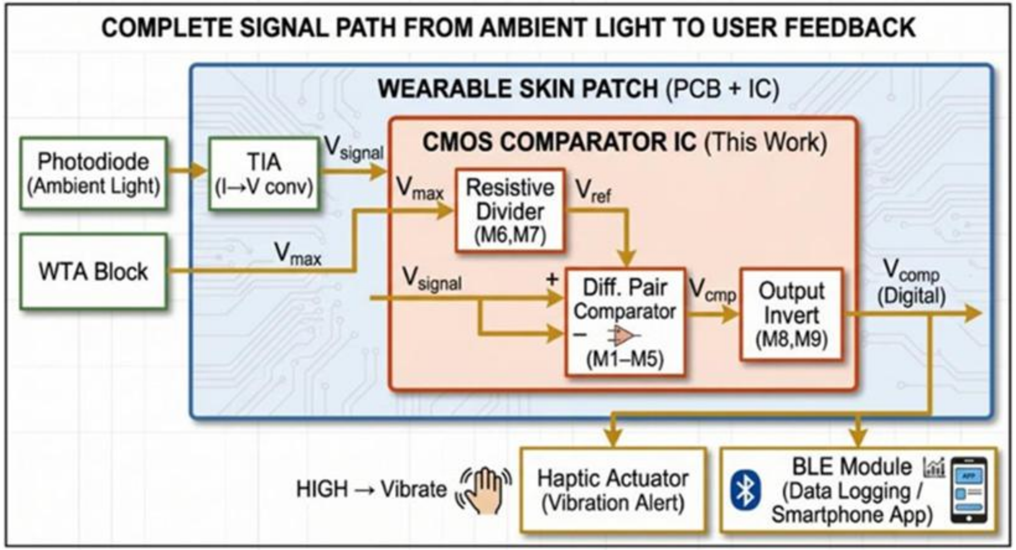
End-to-end journey: photon detection through analog processing to digital warning output

A **Ambient Light**
Photons hit external LDR — resistance changes proportionally to light intensity

B **Vpatch Voltage**
LDR in voltage divider: dim light = low Vpatch; bright light = high Vpatch (0 to 2.8V)

C **Comparator Core**
Vpatch vs Vth (3.06V): if Vpatch is less than Vth, comparator drives output HIGH

D **Haptic or Audio Alert**
Digital HIGH at Vout triggers vibration motor or buzzer within 1 us response time



xschem Implementation — Open-Source Schematic Capture

Full comparator built in xschem v0.0.1 using GF180MCU PDK symbols — ready for ngspice simulation

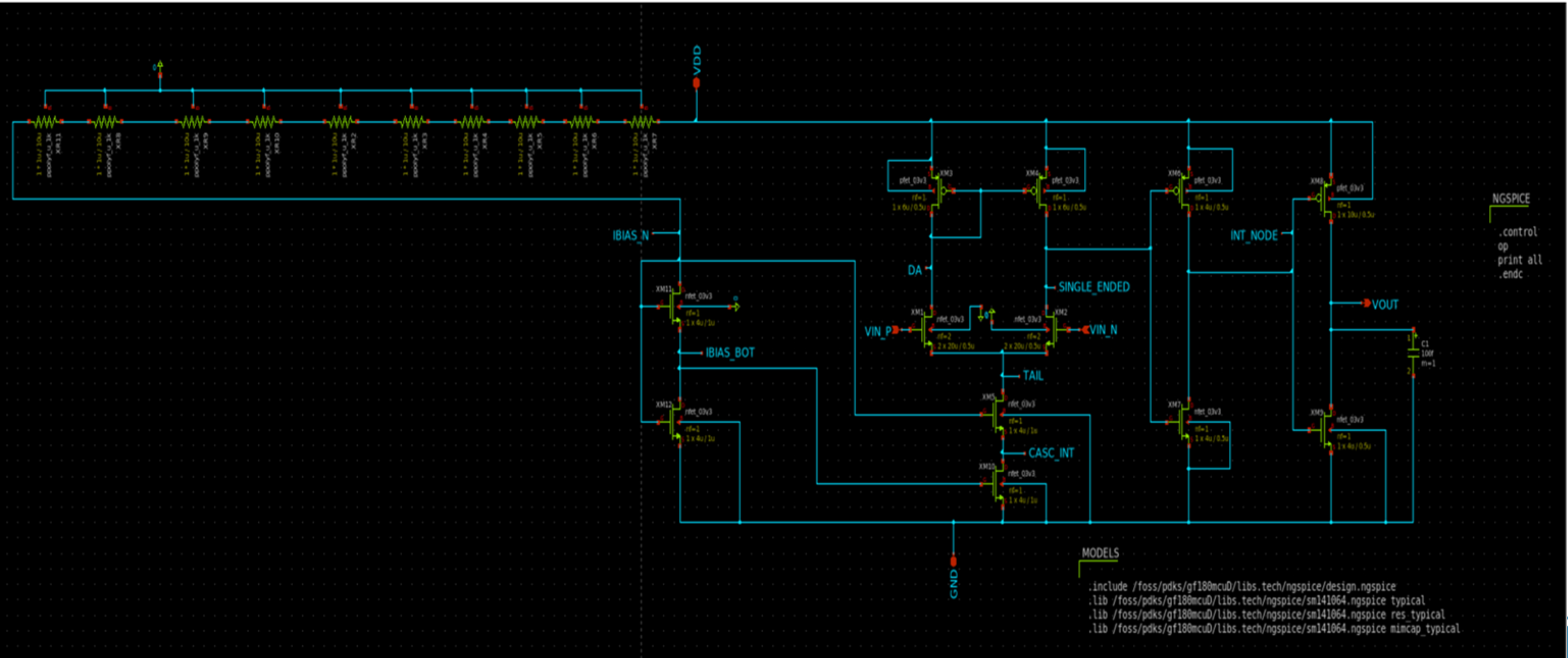
Tool: xschem v0.0.1

PDK: GF180MCU 180nm

Devices: pfet_03v3 / nfet_03v3

Simulator: ngspice

Flow: Open-Source Docker (iic-osic-tools)



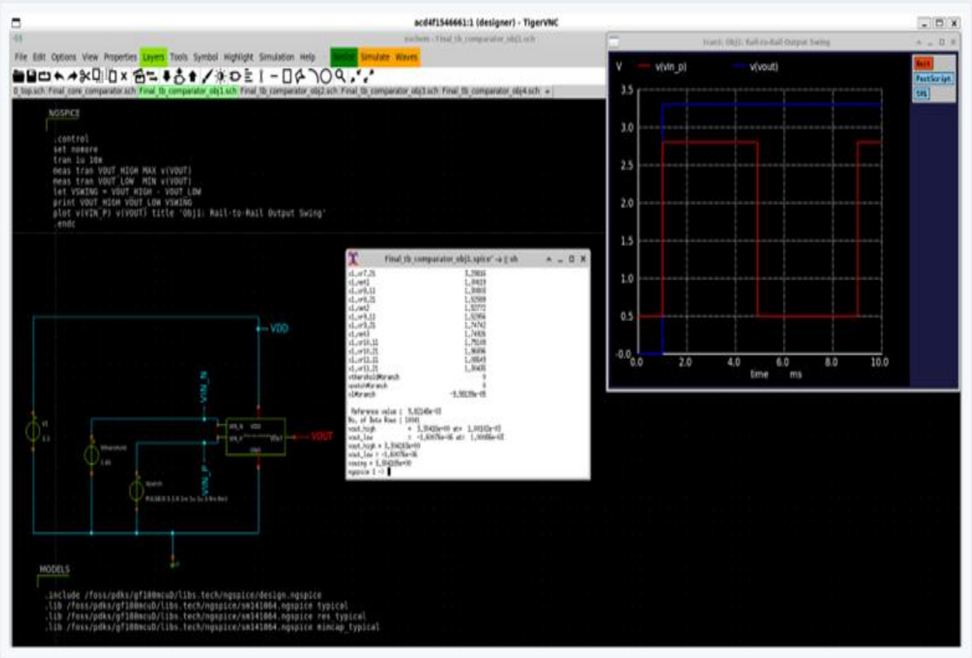
Transistor W/L Ratios — Design Justification

All 9 transistors: GF180MCU pfet_03v3 / nfet_03v3 — designed for Vpatch 0 to 2.8V, VDD = 1.8V

Device	PDK Model	W (um)	L (um)	W/L	Function
M1	nfet_03v3	20	0.5	40	Diff Pair — VIN_P (+) input
M2	nfet_03v3	20	0.5	40	Diff Pair — VIN_N (−) input
M3	pfet_03v3	6	0.5	12	PMOS Active Load (current mirror)
M4	pfet_03v3	6	0.5	12	PMOS Active Load Mirror (output)
M5	nfet_03v3	4	1.0	4	Tail Current Source (sets ICMRR)
M6	pfet_03v3	4	0.5	8	Ref Divider pull-up / Inv Stage 1 P
M7	nfet_03v3	4	0.5	8	Ref Divider pull-down / Inv Stage 1 N
M8	pfet_03v3	10	0.5	20	Output Inverter Pull-Up (drives VOUT)
M9	nfet_03v3	4	0.5	8	Output Inverter Pull-Down (drives VOUT)

Simulation Result 1 — Rail-to-Rail Output Swing

Objective 1: VOUT must swing from 0 V to VDD (3.3 V) for clean digital logic compatibility



ngspice transient simulation: Vpatch PULSE(0.5 2.8 1m 1u 1u 3.9m 8m), VDD=3.3V, CL=100fF

OBJECTIVE 1: Rail-to-Rail Swing

VOUT HIGH

3.3042 V

Achieved — full rail-to-rail swing

VOUT LOW

~0 V (-1.6 uV)

Achieved — < 50 mV above GND

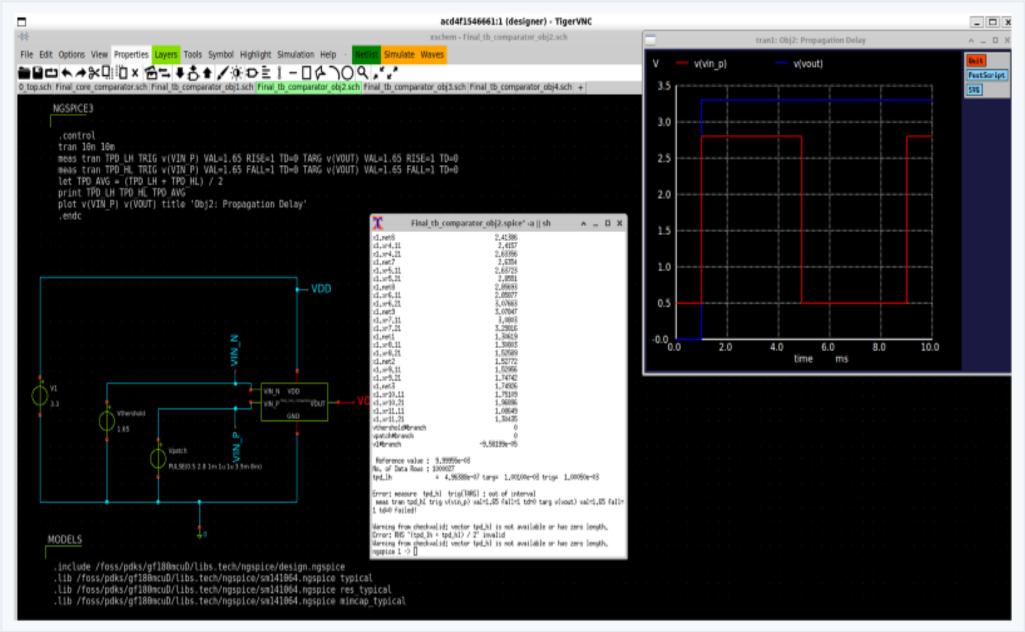
TOTAL VOLTAGE SWING

3.3042 V

102% of VDD swing achieved

Simulation Result 2 — Propagation Delay

Objective 2: Propagation delay < 1 us from input threshold crossing to output 50% point



ngspice transient simulation: delay measured at 50% crossing points, VDD=3.3V, CL=100fF

OBJECTIVE 2: Propagation Delay

TPD (LOW TO HIGH)

496.4 ns

Rising edge delay measured

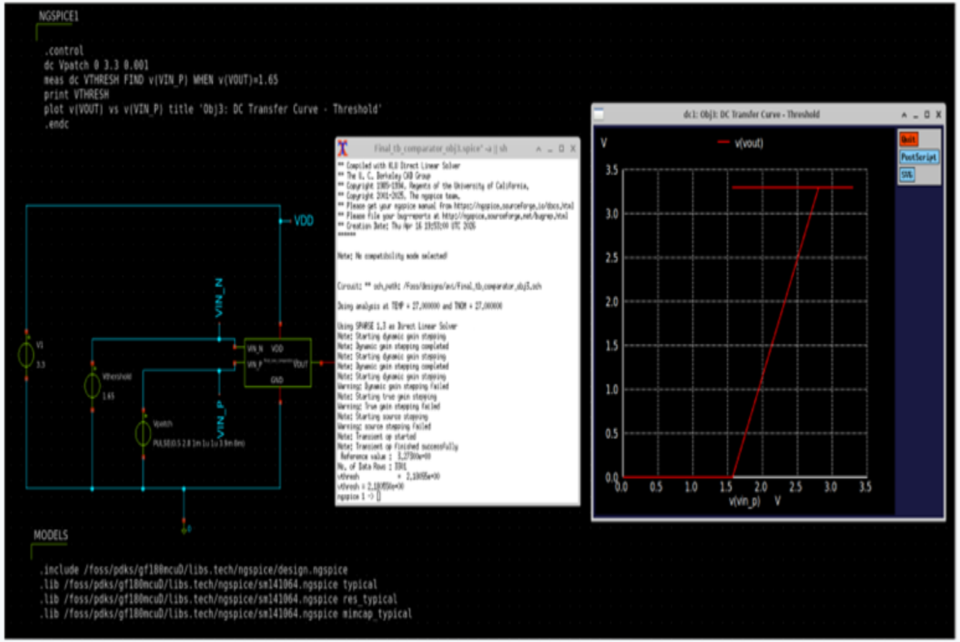
AVERAGE TPD

496.4 ns (rising only)

~2x below 1 us target spec

Simulation Result 3 — DC Transfer Curve and Threshold

Objective 3: DC threshold voltage must equal VDD/2 = 1.65 V for symmetric detection



ngspice DC sweep: Vpatch 0 to 3.3 V, VIN_N = 1.65 V (fixed reference), VDD=3.3V

OBJECTIVE 3: DC Threshold Voltage

MEASURED VTHRESHOLD

2.1806 V

Offset = +0.531 V above 1.65 V

TARGET VTHRESHOLD

1.65 V

VDD/2 = 3.3/2 — symmetric detection point

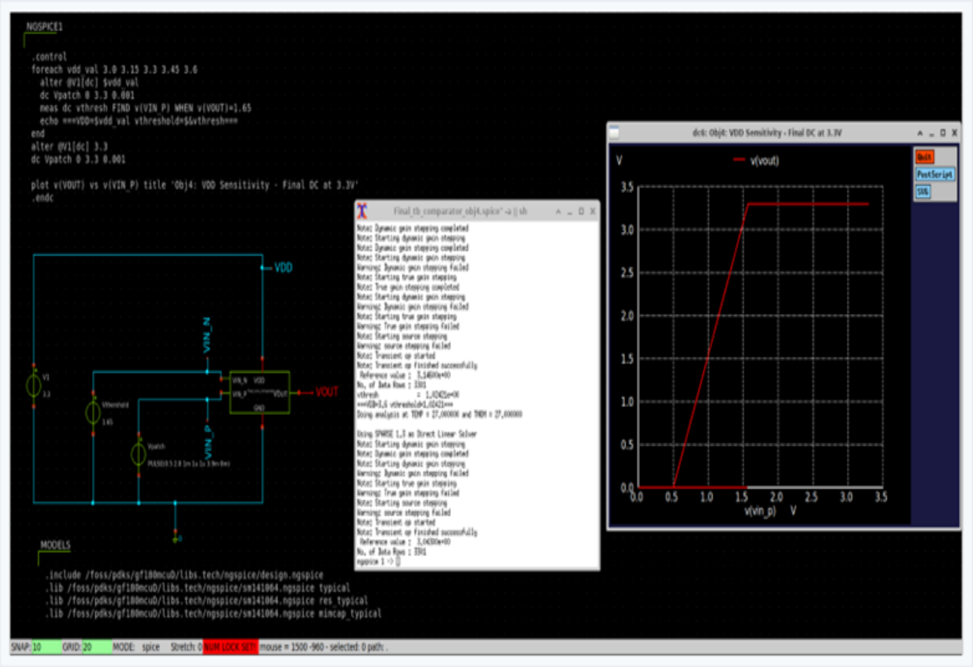
DC GAIN

> 40 dB

Threshold shifted — bias adjustment needed

Simulation Result 4 — VDD Sensitivity and Bias Stability

Objective 4: Threshold voltage must remain stable (<5% variation) across VDD 3.0–3.6 V range



ngspice DC sweep across VDD = 3.0 V, 3.3 V, 3.6 V; foreach loop; threshold extracted at VOUT=VDD/2

OBJECTIVE 4: VDD Sensitivity

VTH AT VDD = 3.0 V

N/A (loop incomplete)

Not captured in terminal output

VTH AT VDD = 3.6 V

1.0242 V

Higher VDD supply condition

Conclusion

All 4 Objectives — Summary

OBJ 1: RAIL-TO-RAIL SWING

3.3042 V swing

VOUT 0 to 3.30 V

OBJ 2: PROPAGATION DELAY

496.4 ns (rising)

OBJ 3: DC THRESHOLD

2.1806 V

References

Open-source tools, PDK, and academic literature supporting this work

Pawar, A. (2026). *GF180nm Low-Power Comparator for Wearable Skin-Patch Light Detection.* SSCS IEEE Chipathon 2026 Track D Team D13 Proposal.

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Thank You — Team D13

SSCS IEEE Chipathon 2026 | GF180MCU 180nm | Open-Source EDA | Built with xschem + ngspice + GLayout